Kartik Somani

Bengaluru Relocation Ready | +91 63779 37036 | kartik.somani500@gmail.com | kartikdesign.vercel.app | github.com/ktsomani | linkedin.com/in/kartik-somani

Professional Summary

High-agency **Design Engineer** with a B.Tech in Electronics and Communication Engineering (ECE), combining hardware-level systems rigor with interface craft, 60fps motion choreography, and creative browser rendering. Deeply focused on non-linear editing (NLE) interfaces, real-time multiplayer states, and multimodal AI interaction models. Leverages an AI-native design engineering workflow (directing Claude for logic/state exploration and Figma for design systems) to prototype, build, and ship production-grade software using React, Tailwind CSS, Framer Motion, and WebGL/GLSL.

Education & Certifications

Bachelor of Technology (B.Tech) – Electronics and Communication Engineering (ECE) *Graduated 2024*

Primavera P6 Fundamentals (Certified) – Oracle Primavera P6 (WBS, Critical Path Method) *Alison*

Technical Skills & Design Engineering Stack

Motion, Animation & Creative Code: Framer Motion (Layout Animations, Gesture Drag, Spring Physics Tuning), React Three Fiber (R3F), GLSL Fragment/Vertex Shaders, WebGL Pipelines, Canvas 2D API, SVG Choreography, Micro-Interactions

Frontend Architecture & Craft: React, Next.js, TypeScript, JavaScript (ES6+), Tailwind CSS, Component Architecture, State Management, DOM Virtualization, 60fps Frame Budgeting, Memory Profiling, Responsive Web Layouts

Design Systems & Curation: Figma (Variables, Tokens, Auto-Layout, Interactive Prototyping), Interface Craft Guidelines, Dark Mode Ergonomics, Typographic Ratios, WCAG 2.1 AA Accessibility, Prompt Engineering (Claude, Midjourney)

Backend & Developer Tooling: Node.js, Python, SQL (PostgreSQL, SQLite), RESTful APIs, Git/GitHub, Docker Basics, Postman, Vite

Work Experience

Jaipur Metro Rail Corporation (JMRC) Jaipur, India

Engineering Summer Intern *June 2023 – August 2023*

- Audited high-density telemetry displays and operational interface logs, isolating visual clutter that caused operator cognitive fatigue during alert triage.
- Redesigned complex diagnostic inspection checklists into standardized visual SOPs, improving information hierarchy and verification throughput.

Bosch India

Engineering Trainee (Summer Training) *June 2022 – August 2022*

- Analyzed electronic test bench telemetry feeds, verifying sensor signal tolerances against technical hardware specifications.
- Engineered automated spreadsheet inspection routines to flag anomalous readings and signal drift, adhering to strict industrial quality benchmarks.

Cardboard-Targeted Design Engineering Projects

Collaborative Timeline Editor (Browser NLE) *React, Tailwind CSS, Framer Motion* | [Live Demo]

- Engineered a browser-based non-linear video editing (NLE) timeline featuring gesture-based clip dragging, dynamic track duration trimming, and magnetic grid snapping.
- Built a real-time multiplayer simulation rendering a secondary user's interpolated cursor presence and concurrent track modifications to evaluate multi-user ergonomics.
- Optimized React re-render lifecycles using targeted memoization and CSS transform offloading, preserving a stable 60fps interaction loop during rapid scrub operations.

Multimodal AI Video Search & Context Dashboard *React, Tailwind CSS, Framer Motion* | [Prototype]

- Designed an intent-driven search surface translating natural language scene descriptions into frame-accurate video segment results with fluid layout transitions.
- Implemented a slide-out "Context Panel" animating the model's underlying reasoning graphs, confidence scores, and localized temporal timestamps.
- Orchestrated layout enter/exit transitions with Framer Motion layout animations to preserve user spatial orientation across search query refinements.

WebGL / GLSL Motion Graphics Studio *React Three Fiber (R3F), GLSL, Three.js, Framer Motion* | [Demo]

- Built an in-browser creative tool rendering procedural 3D geometries powered by custom GLSL fragment shaders (dithering and chromatic glitch displacement).
- Exposed reactive parameter controls for color curves, rotation velocities, and shader uniforms without blocking the WebGL rendering thread.
- Synchronized 3D canvas viewport transformations with 2D DOM controls using spring physics, demonstrating high-performance web graphics execution.